OH MY CODE (OMC) PROGRAMMING COMPETITION



PROJECT DETAILS

PROJECT TITLE: PARCEL DELIVERY SYSTEM

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ORGANIZATION OVERVIEW

Group Name: DeliverUs

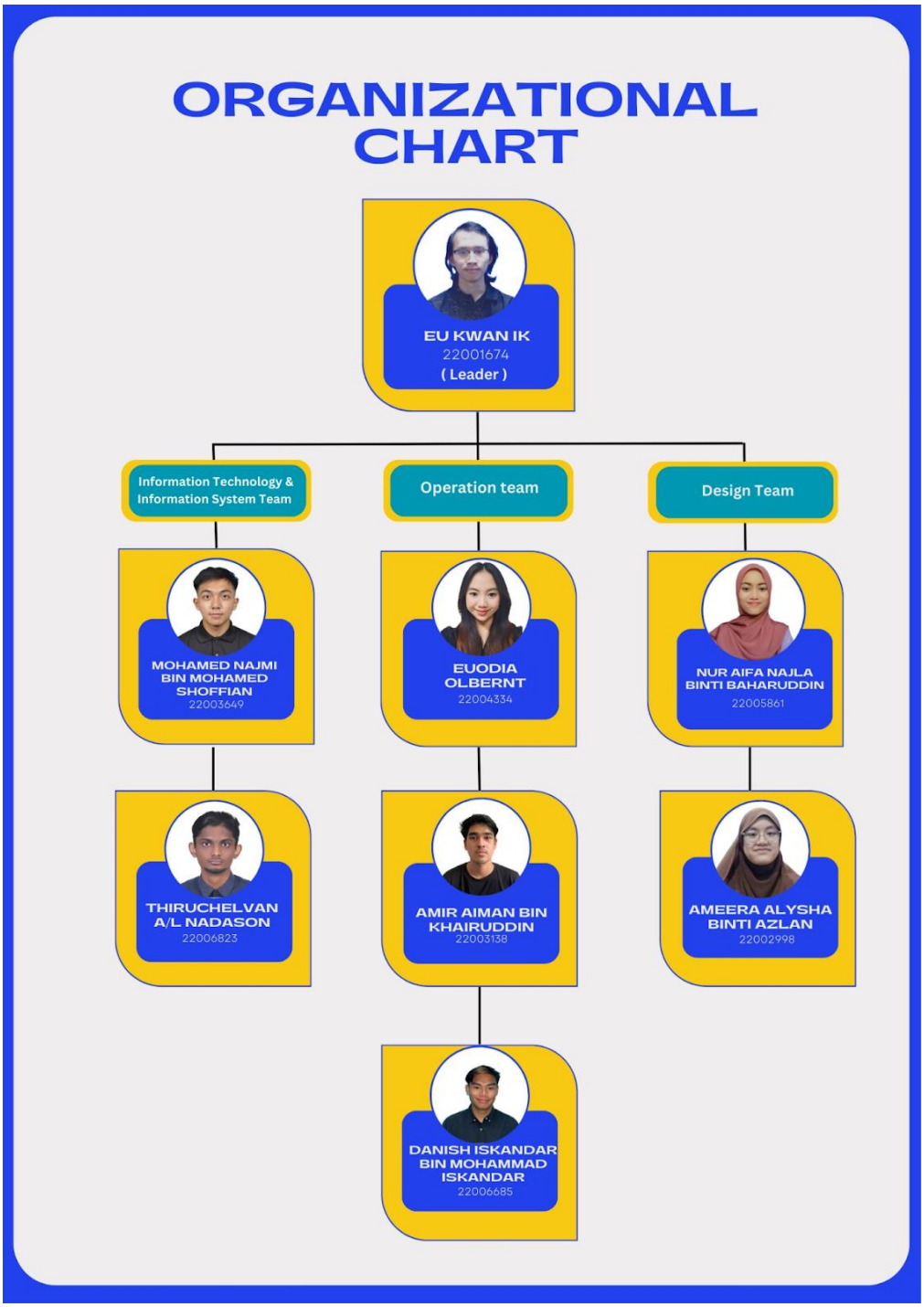
Group Logo:



- Team member details (Name, Student ID, Program)

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3.	Mohamed Najmi Bin Mohamed Shoffian	22003649	Computer Science
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- Organization chart



1. BACKGROUND OF STUDIES

The UTP parcel system was created in 2006 to allow students to purchase items online and have it delivered straight to the campus. (Norbaya Aida binti Nordin, June 2006, Internal Parcel Management System (IPMS), pg. 2-3) The parcel hub, originally named “Lembaran Residential office”, was where all of the parcels ordered by UTP staff or students are situated at. The receiver would have to travel from their residential area to the office to check whether their parcel had arrived. However, this system is flawed as it does not have any notification system to notify that the parcel has arrived due to the limited technology in 2006. Thus, students may waste their time and energy having to travel to the office only to find that their parcel hasn’t arrived. Furthermore, parcels that are neglected or have not been collected within three days of arrival would be sent back to its sender. This can trouble the recipient of the parcel as they may have other things to do, and would have no time to collect their parcel, which further highlights the importance of a parcel being delivered conveniently and on time.

This issue had been voiced out by the SRC (Student Representative Council) in UTP, and a new and more reliable parcel collection system had been established since 2018. The parcel hub system was relocated to Village 4 mart which is a well-suited location for students residing in Village 3, Village 4 and Village 5. Additionally, recipients would be notified when their parcel has arrived through their emails due to improved technology. Students could also check the status of whether the parcel has arrived or not through the SRC UTP website. During the Covid-19 pandemic, UTP’s Covid Response Team appointed Pos Malaysia Berhad as the service provider for UTP Corporate Mail management and students must use their service to receive and deliver parcels on campus during the Conditional Movement Control Order. As of 2023, the parcel hub has been relocated to Block M next to Maybank.

Students in UTP are obliged to stay at campus as part of the requirement to enrol. Therefore, students will have to buy their own essential needs. Most students prefer ordering online. This is because most online platforms offer a huge selection and variety of products due to physical stores in Seri Iskandar having limited stock. Shopping online is also convenient due to its functionality to purchase an item without being present in stores. It also offers better prices and discounts depending on the season, such as the well renowned shopping online platform Shopee, which offers free delivery and discounted products every month. Therefore, students would order items online, whether it is their wants or needs, and have it delivered straight to the campus.

Currently, the parcel system at UTP operates through a centralised parcel hub. When students purchase packages, they are sent to this hub, where they are sorted before becoming available for self-collection. However, this system poses various challenges for students. Firstly, it can be quite burdensome as students may lack the time or energy to pick up their parcels from the hub. Moreover, some students' accommodations are located far away from the hub, making it inconvenient for them to access it, especially if they lack transportation options. Additionally, students are required to pay a sorting fee before they can retrieve their parcels, adding to their financial burden.

During their academic journey, students may encounter financial difficulties. Tuition fees, textbooks, and other educational materials could cause a burden on students. Besides that, the cost of living including food, transportation, and entertainment can further exacerbate their financial issues. In relation to this, a common problem students face is managing their expenses. Due to their lack of experience in managing finances by themselves, some students may participate in unnecessary spending. To address this problem, the students find themselves requiring a source of income to support their daily lives. One of the most common solutions is pursuing part-time jobs. Many students choose to work part-time while studying to earn an income. Universities and local businesses frequently offer part-time positions specifically designed to accommodate students’ schedules. These jobs not only provide much-needed financial stability, but also offer valuable work experience that can enhance students' future career prospects.

In correlation with the OMC project theme, which is digital transformation, the definition of the integration of digital technology into all areas of a business, fundamentally changes how you operate and deliver value to customers. It is also a cultural change that requires organisations to continuously challenge the status quo, experiment, and get comfortable with failure. Traditional jobs like sales, marketing, and customer service are transcended by digital transformation. With digital technology on our side, we have the ability to rethink how we conduct business and interact with our customers as we progress from using paper, spreadsheets and smart applications to manage our enterprises.

In relation to the matters we have discussed, the parcel system in UTP has a long history and a massive redevelopment phase due to technology constantly changing. However, its shortcomings are still visible to this day.

2. PROBLEM STATEMENT & OBJECTIVES

Problem Statement

Since our university's parcel collection has changed from V4 Mart to the parcel hub beside Maybank, a majority of UTP's students have complained about the parcel hub being too far from their respective villages. Some students who do not have any kind of transportation are facing hardships collecting their parcels. Students that are most affected are those that live in Villages 3, 4, 5, and 6, as their villages are located very far from the parcel hub. This shows that the relocation of the parcel hub from V4 Mart to its new location causes a lot of inconvenience.

Furthermore, the relocation of the parcel hub has had a significant impact on students, particularly in terms of time and energy. It takes a lot of time and effort to line up in a long line to collect parcels. Moreover, the parcel hub at Block M also faces the problem of limited space, so students will have to queue outside the hub if there is no room. This problem is due to the new operational schedule, as students are very likely to collect their parcels after they finish their classes at Block I or Block J. Consequently, most students choose not to collect their parcels at other times, as it would require them to take a significant amount of time to travel from their village to the parcel hub, especially those without access to transportation, particularly foundation students.

As of lately, many students are facing financial problems as the cost of living is on the rise. Whether it is habitual or compulsive spending, students are looking for ways to earn extra pocket money for their daily expenses. They are working hard to find ways to earn extra money for their financial needs. Students seek part-time jobs on campus due to their convenience, even if they have busy schedules.

Objectives

The goal of our program is to incorporate digital transformation aspects in UTP's society. We aim to increase the power of digital technology to transform the traditional parcel delivery practices and enable UTP students to adapt to the rapidly changing digital landscape. It involves not just the implementation of technology, but also the reconstruction of business strategies, organisational structures, and customer interactions.

Our primary objective in the making of the program is to provide a convenient and reliable way for students to receive their packages on campus. Students at UTP can receive their packages without any inconveniences or delays through the automation of processes. Our system is designed to improve operational efficiency, reduce costs, and minimise errors, as well as reduce the likelihood of delays to ensure the smooth and timely movement of parcels in the delivery process.

Concerns have been raised over students struggling at universities in the country due to problems of affordability, added with the rising cost of living. (Isya Zulkifli, 29 September 2022, HOW INFLATION AFFECTS MALAYSIAN UNIVERSITY STUDENTS, https://cilisos.my). Therefore, students are looking for a source of income through part-time jobs or entrepreneurship. Fortunately, our system is designed to create opportunities for UTP students who want to earn some extra money by delivering parcels as couriers. This system is an option for UTP students who want to earn an income while balancing their studies and other commitments. By using our system, UTP students can earn extra income without the need for a traditional part-time job. With our user-friendly platform, students can easily sign up and earn money quickly. Other than that, students will have the opportunity to gain valuable work experience as couriers and develop their skills in areas such as time management, communication, and customer service.

Furthermore, the purpose of implementing a parcel delivery system in our university is to streamline the management and delivery of parcels. As we all know, the current system of parcel collection at the parcel hub near Maybank has caused inconvenience for students, particularly those located in Villages 4, 5, and 6, who find it difficult to travel to the hub due to distance and a lack of transportation. We intend to simplify the package management procedure for students by introducing a user-friendly system. This features a centralised platform where students can quickly register items, change their delivery settings, and receive notifications.

In addition, the program we run is able to help with reducing parcel loss or misplacement, which will reduce the possibility of parcels becoming lost or forgotten during the collection process. The university's focused delivery mechanism ensures timely and correct distribution to the intended recipients. This allows students to create trust and alleviate concerns about the safety and security of their deliveries.

Last but not the least, our company aims to deliver customer satisfaction through data analytics. With the consent of our customers, we collect, analyse and derive insights from vast amounts of data to make informed decisions. Users can write their feedback after using our services to rate their satisfaction. This can enable us to personalise and enhance customer experiences.

3. PROJECT DESCRIPTION

Register

On the registration interface is the user interface component that allows our customers to create an account or register for our services. It provides an option for users to choose between whether they want to register as a customer.

Next when the user has selected their choice, they will have to sign up an account to use our services. They will be required to provide relevant information such as username, email and password for account security. However, if they already have an account, they will have to log in into their respective account simply by entering their username or email address and the correct password for the account. If the user forgets their password there is an option to reset their password directly sent to their email. This feature allows users to regain access to their accounts when they have forgotten their password or need to reset it for security reasons.

Courier

The purpose of the courier main menu is for our user to access their profile, courier status and wallet. Couriers can update any relevant information about themselves in the profile section. They can also check on available parcels that are ready to be delivered from the forms submitted by our customers.

Customer

The user profile function is for users to submit a form that contains their parcel details such as tracking number, size of parcel, etc. Users won't have to provide their name, contact information, preferred method of delivery, and address book as it will be passed on automatically to our courier immediately.

The payment option menu is for users to choose their payment choices. The available payment option is either QR Pay or Cash on Delivery (COD). Through the quick response (QR) payment method, customers can easily transfer the money easily. Customers who prefer to pay through COD can pay directly to the courier who delivered their parcel. DeliverUS makes it simple for consumers to pay for their delivery without fearing that their financial information is being compromised by providing a range of payment methods and security safeguards.

Under the check parcel function, users can choose to view the status of their parcel or choose to submit a new form to request a parcel pickup service from us. Furthermore, users can provide feedback and ratings for the delivery service anytime. They can also launch a report or complaint regarding their experience with DeliverUs.

Admin

In the admin menu features such as customer and courier explorer allow administrators to search and see customer and courier profiles which contain important information including client data, order history, shipping preferences, etc. With this, administrators are able to effectively manage customer and courier accounts, update data as needed, and access pertinent information for analytics and reporting. Other than that, administrators can examine and validate the information provided by clients and couriers. This protects against fraud and assures the validity of user accounts. The authorization procedure aids in ensuring that only legitimate users are given access to the system.

Administrators can also view the feedback management module where reports from our users. Administrators may read and assess client evaluations, ratings, and comments using the module to learn more about the calibre of the couriers' services and pinpoint areas in need of improvement.

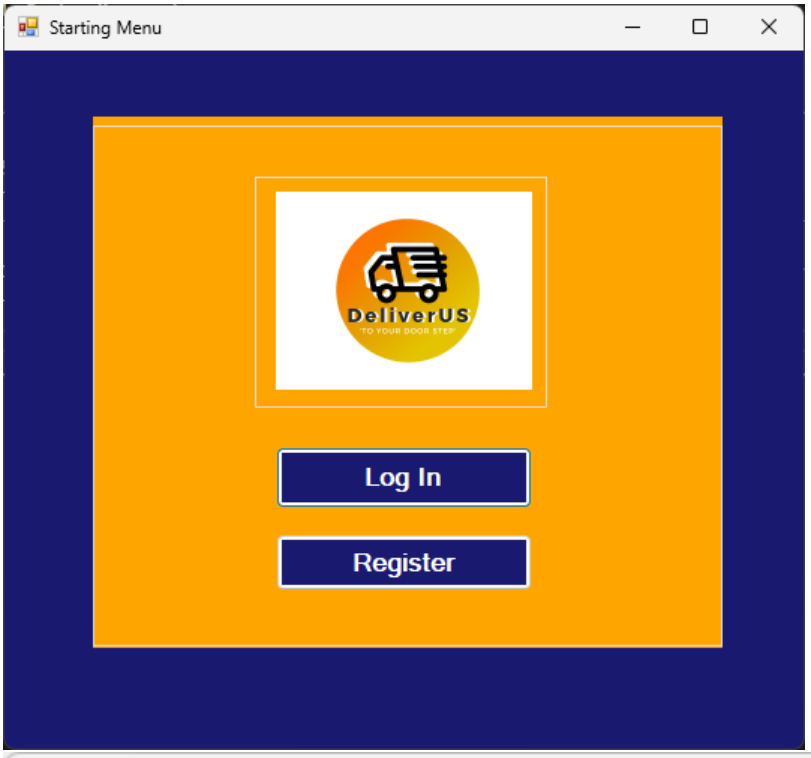
4. CONTRIBUTION TO MANKIND

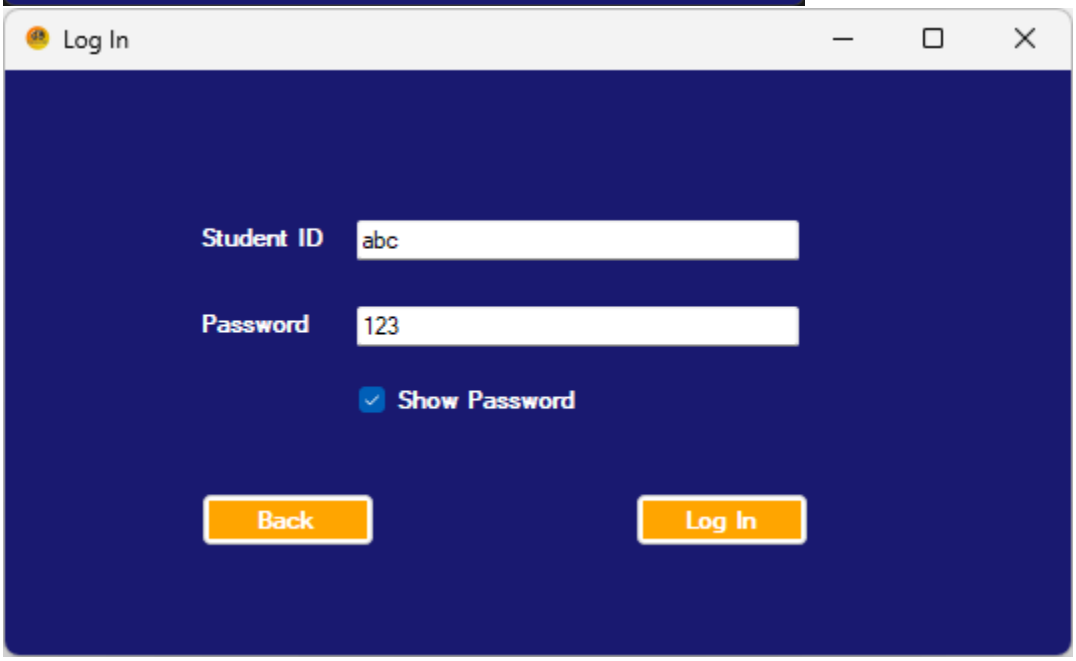
Following the theme of the OMC project, our program will digitalize the parcel delivery system in UTP. With the use of our program, students will be able to easily receive their parcels without the hassle of travelling to the parcel hub located in block M. Our service will solely consist of UTP students who are familiar with our university's complex and wide layout. They can navigate by walking or through micro mobility such as bicycles or scooters.

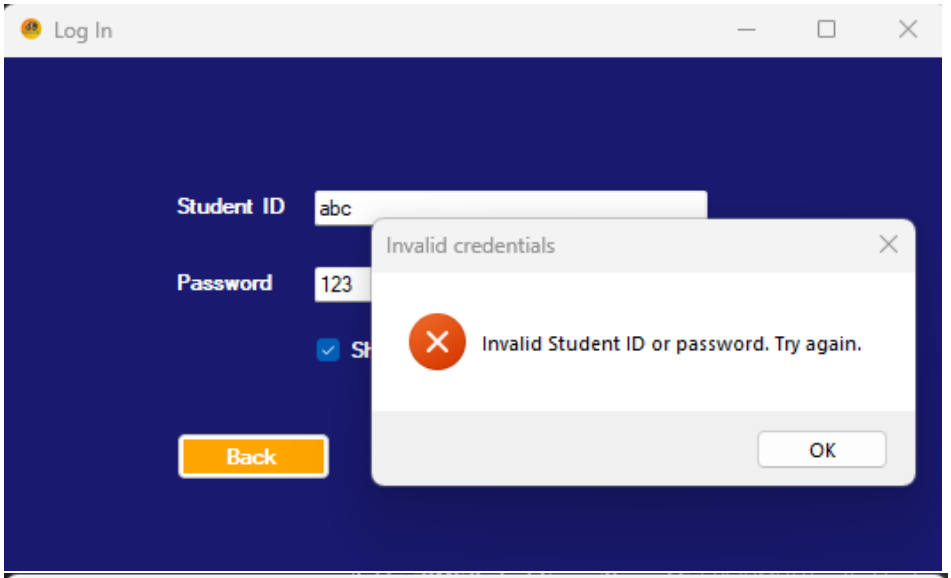
For a majority of university students, their monthly expenses can reach up to RM1000 a month on average (Zulita Mustafa, 8 March 2017, Students on a budget, NewStraitsTimes, https://www.nst.com.my), and students may seek a source income by working a part-time job or starting their own business. Our program can provide students an easy way to earn side income by delivering parcels. Students will earn money depending on the number and size of parcels they have delivered. The more parcels they deliver, the more money they will earn.

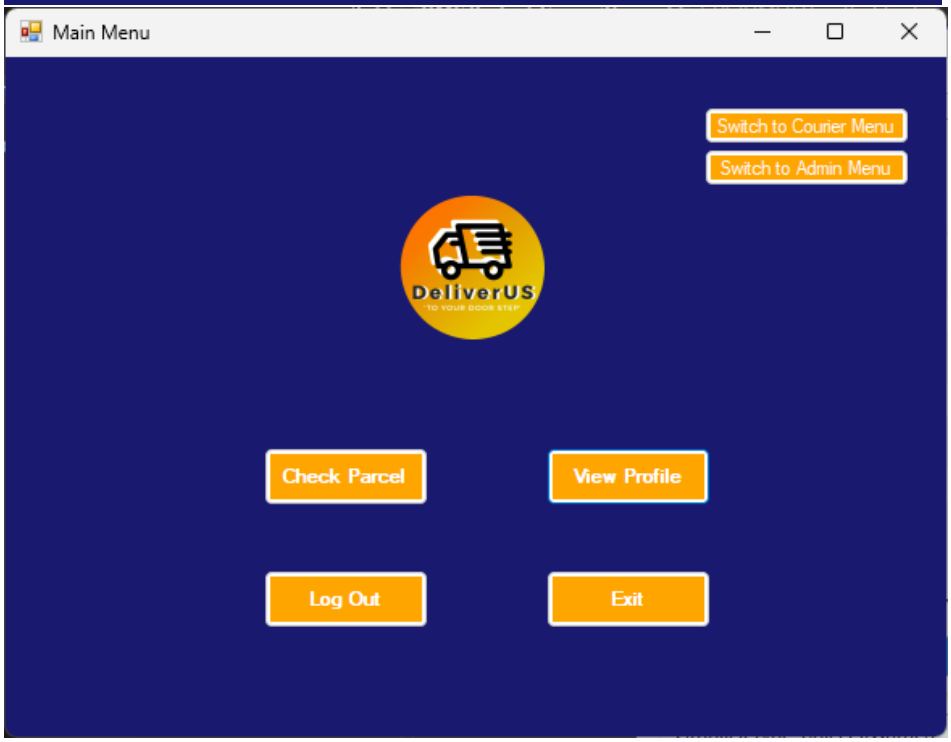
DeliverUs aims to establish a mutually beneficial relationship among customers and couriers. Couriers can expect to complete their tasks with ease with the tools and technology provided to work efficiently using their access to real-time data. Meanwhile, customers will be provided with an easy way to track their parcels and excellent delivery services that will guarantee their satisfaction. Together, with these aspects, our company can adapt to market changes and demands and achieve sustainability and relevancy among UTP society.

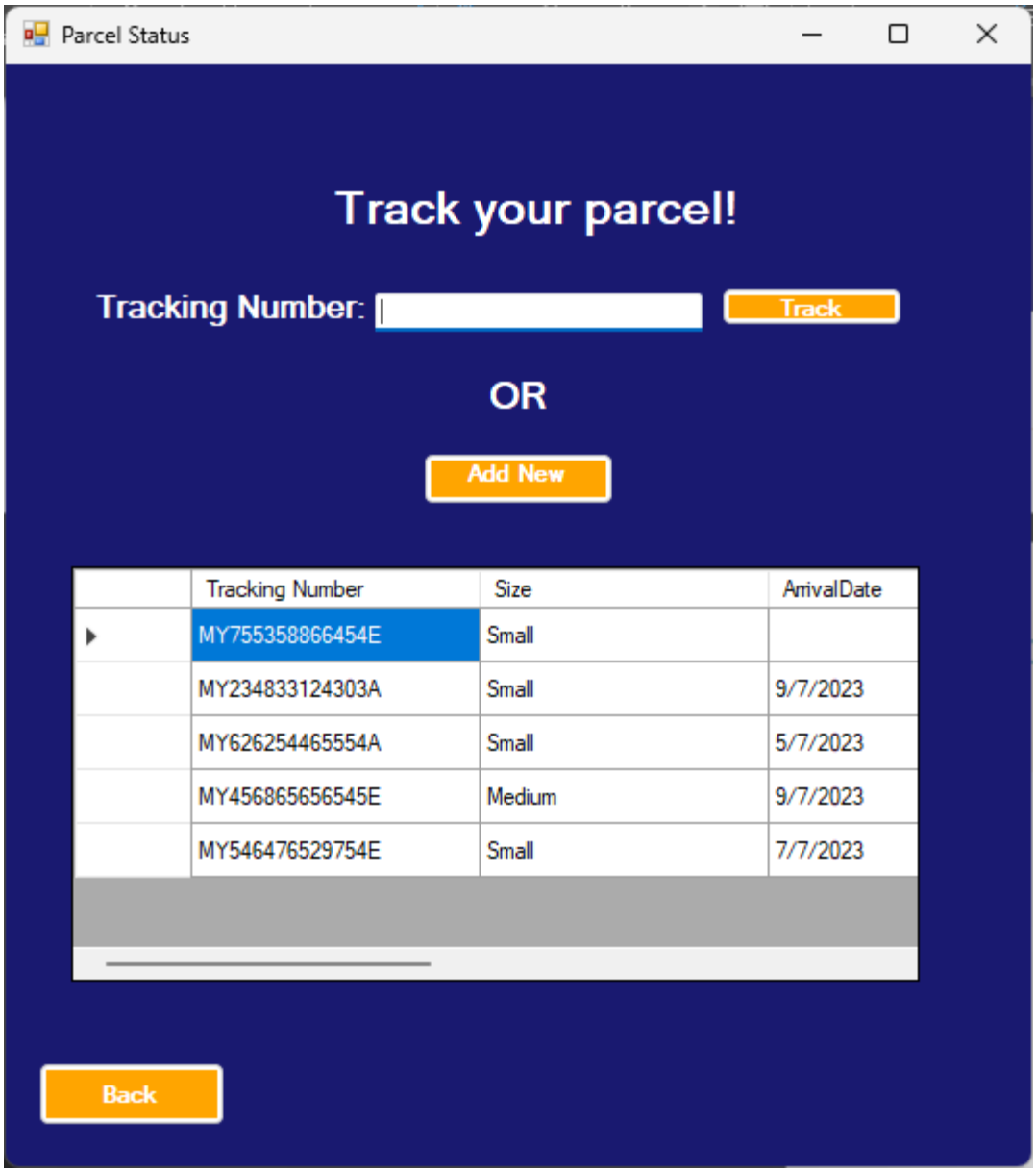
5. APPENDIX - SYSTEM INTERFACE

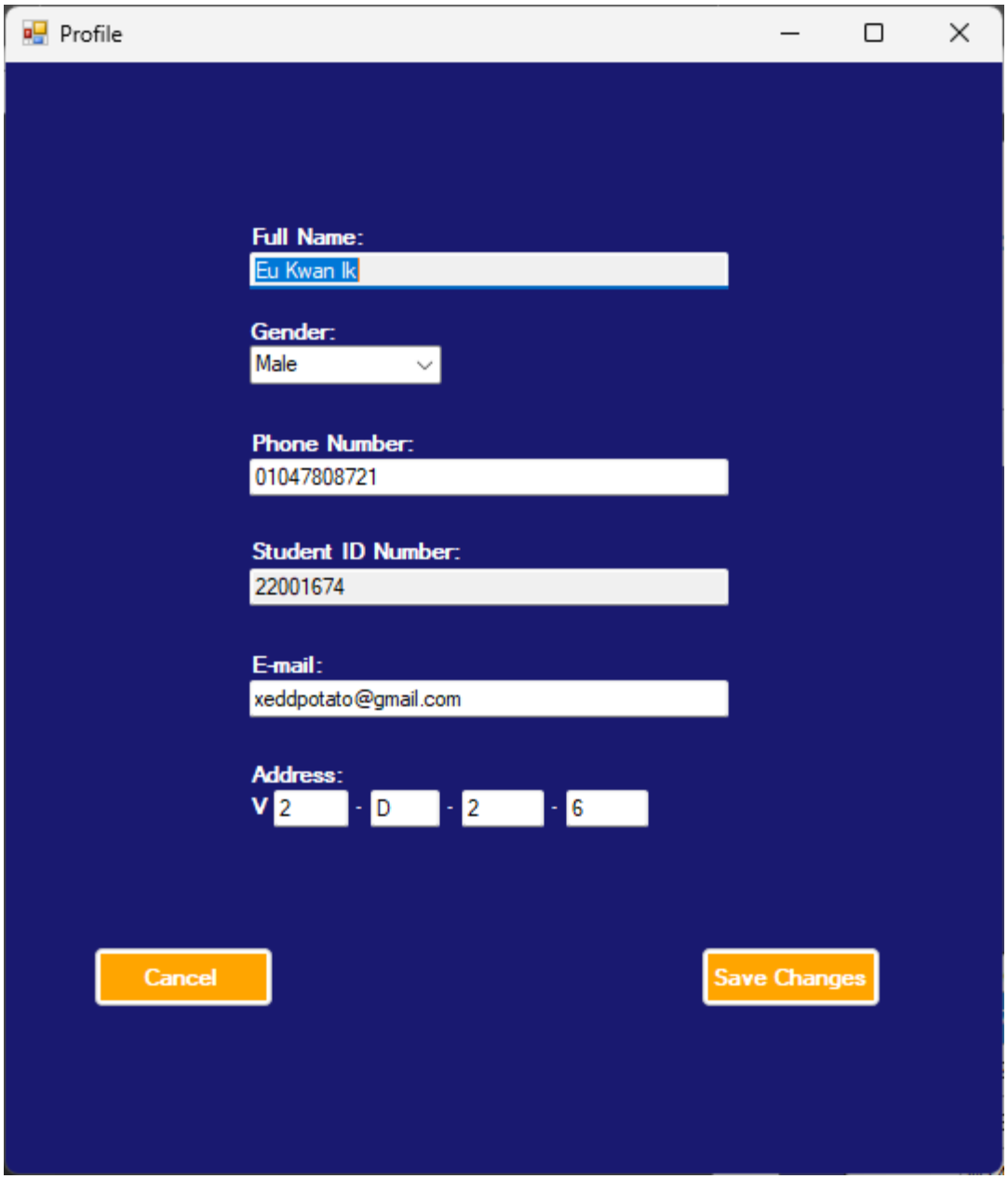


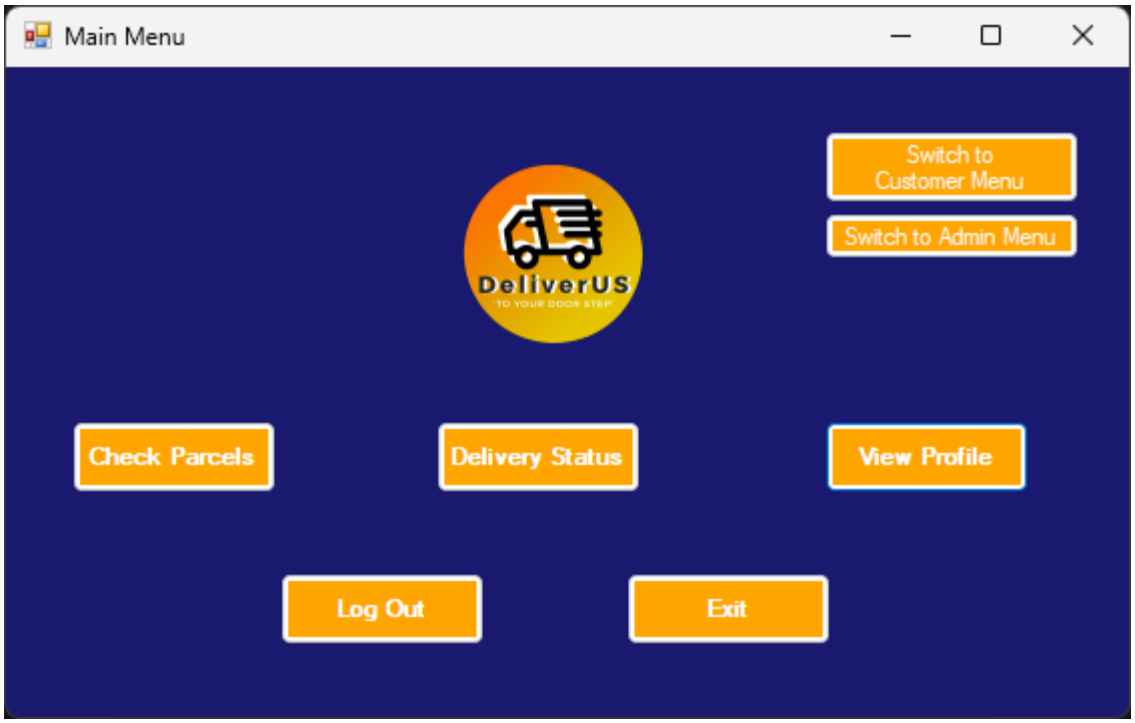


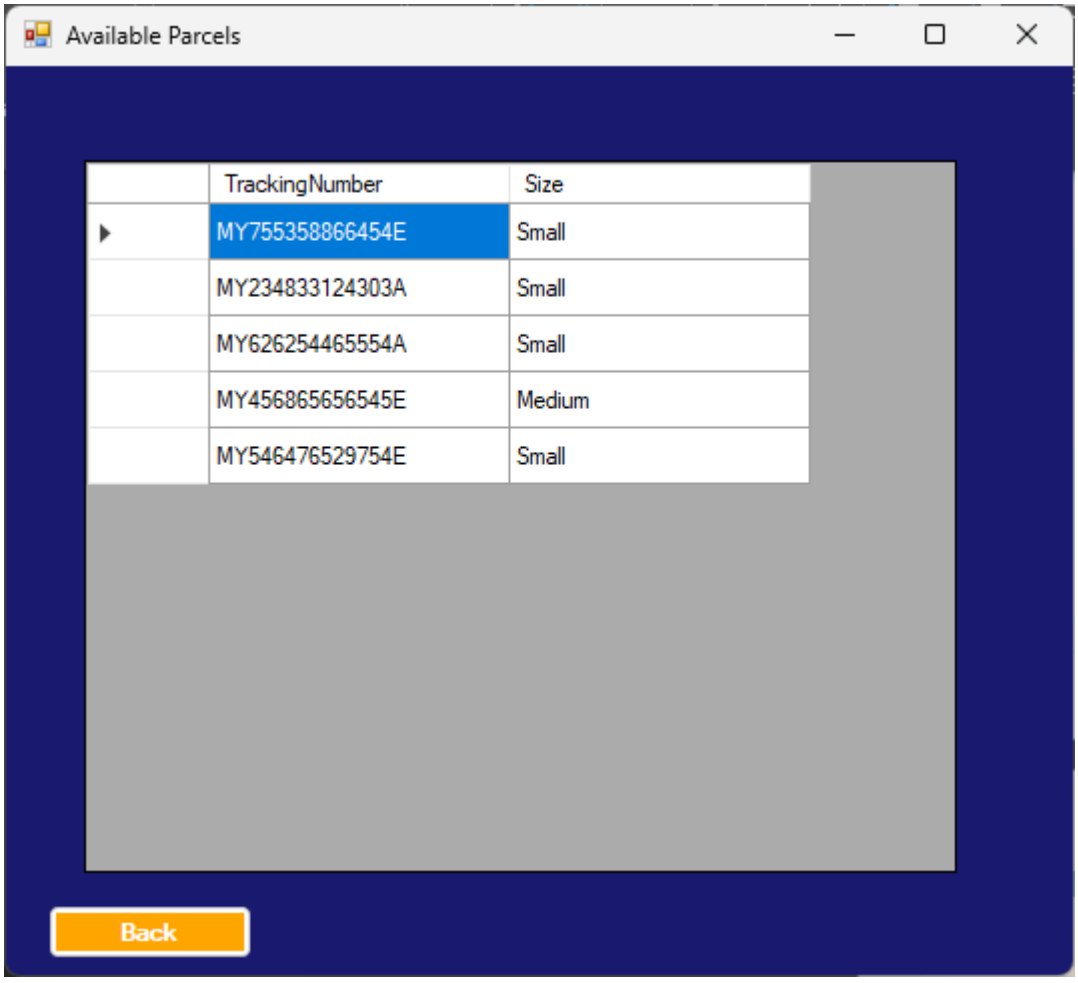

Log In
Student ID
abc
Password
123
Back
Invalid credentials
Invalid Student ID or password. Try again.
OK

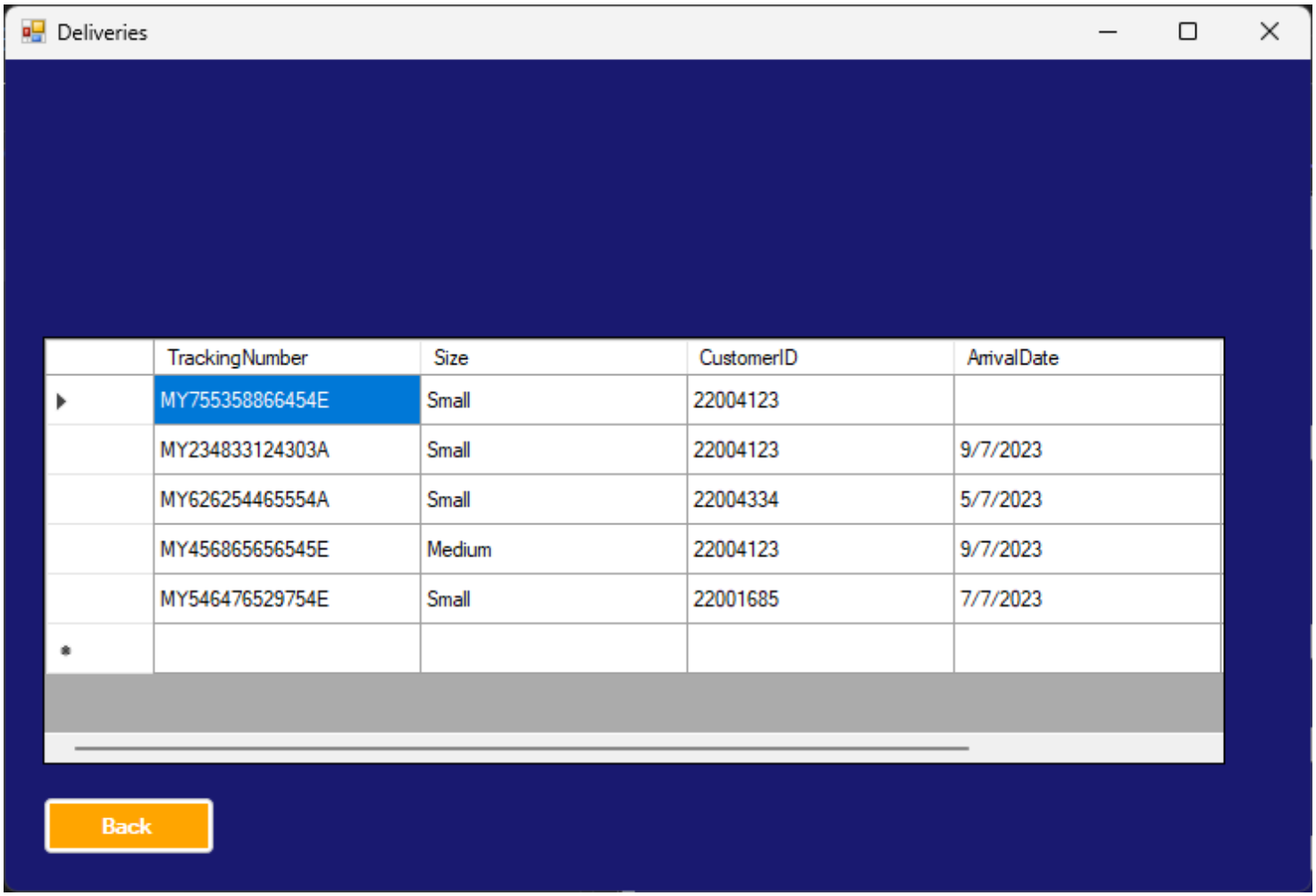

Main Menu
Switch to Courier Menu
Switch to Admin Menu
DeliverUS
Check Parcel
View Profile
Log Out
Exit

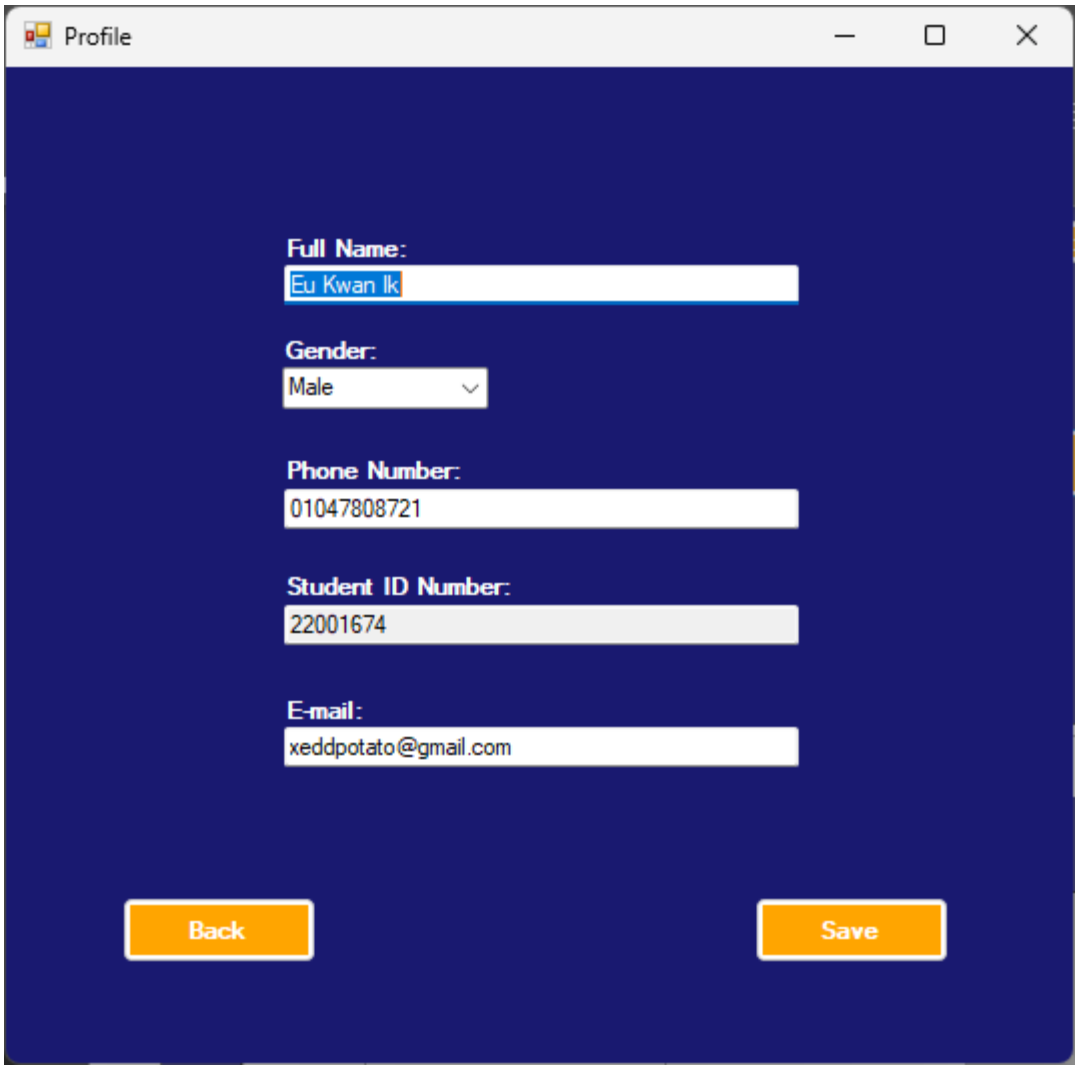

Parcel Status
Track your parcel!
Tracking Number:
Track
OR
Add New
Tracking Number
Size
ArrivalDate
MY755358866454E
Small
MY234833124303A
Small
9/7/2023
MY626254465554A
Small
5/7/2023
MY456865656545E
Medium
9/7/2023
MY546476529754E
Small
7/7/2023
Back

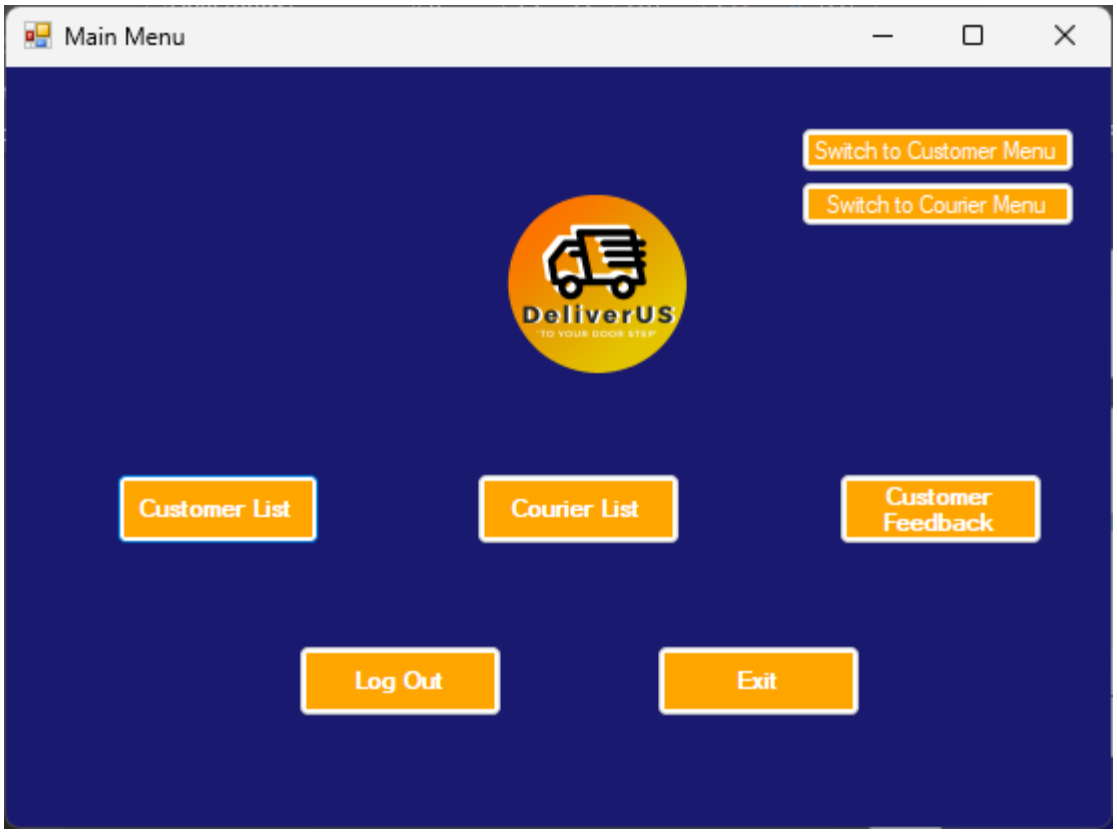

Profile
Full Name:
Eu Kwan Ik
Gender:
Male
Phone Number:
01047808721
Student ID Number:
22001674
E-mail:
xeddpotato@gmail.com
Address:
V 2 - D - 2 - 6
Cancel
Save Changes

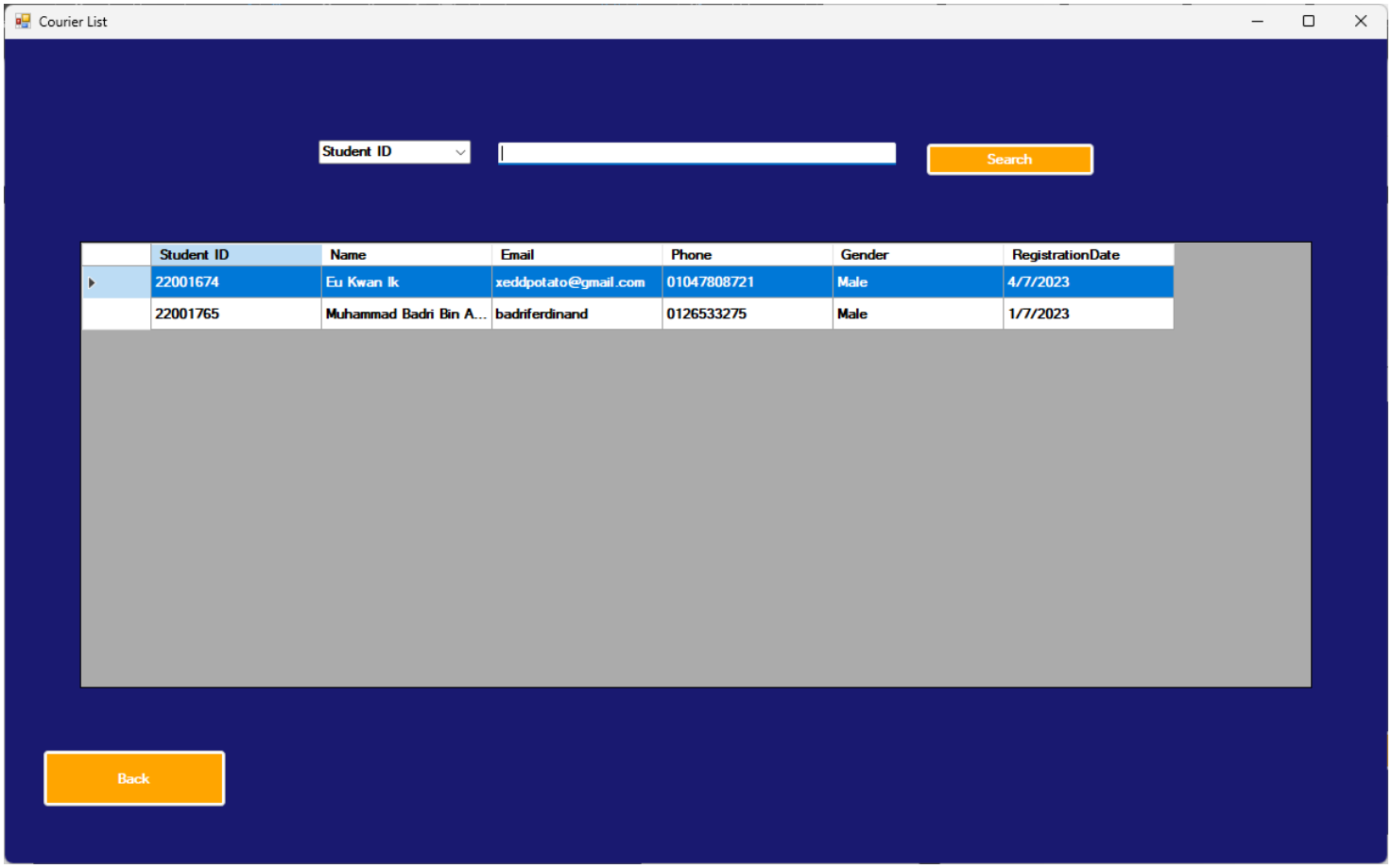

Main Menu
Switch to Customer Menu
Switch to Admin Menu
DeliverUS
Check Parcels
Delivery Status
View Profile
Log Out
Exit

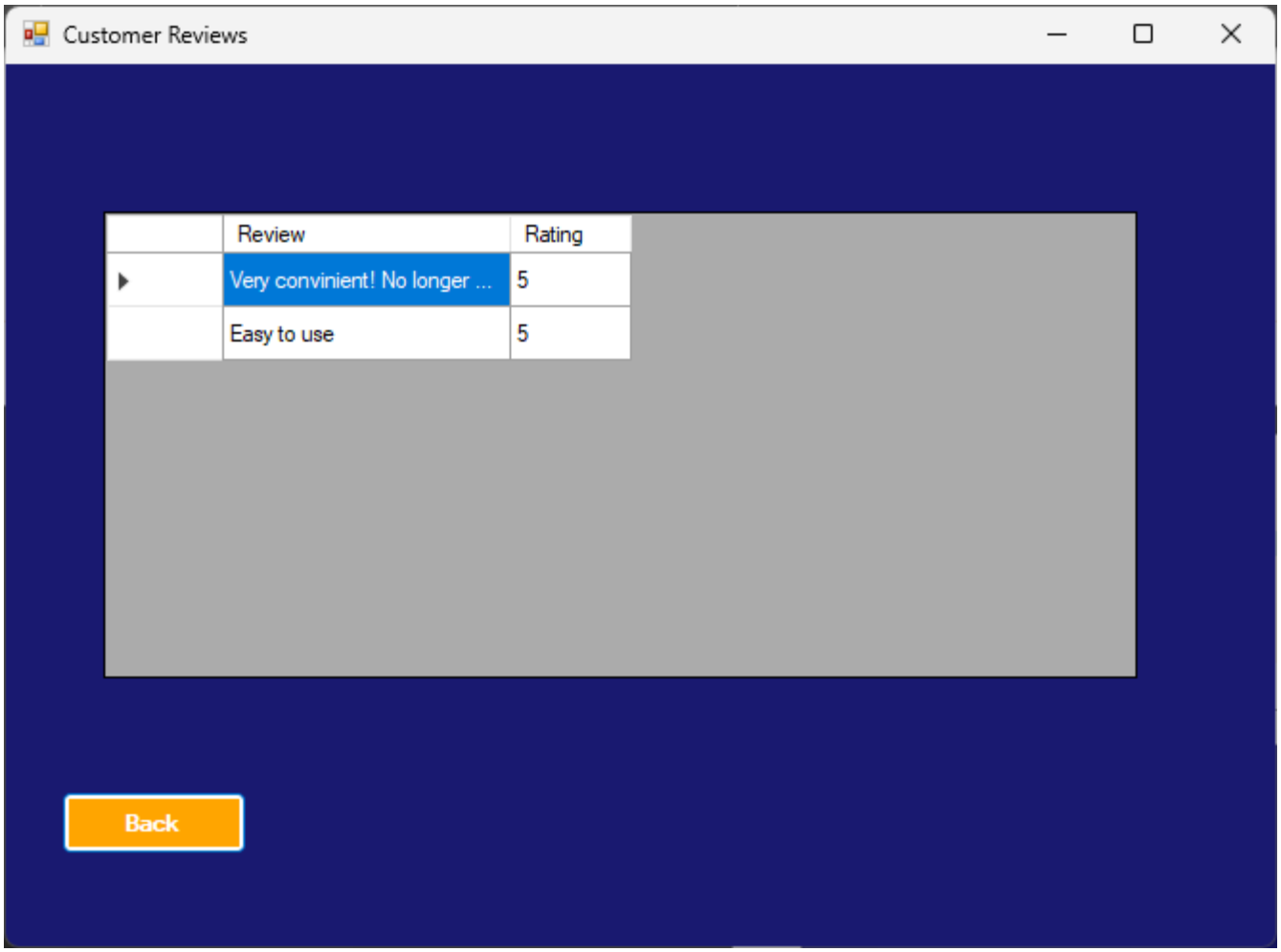

Available Parcels
TrackingNumber
Size
MY755358866454E
Small
MY234833124303A
Small
MY626254465554A
Small
MY456865656545E
Medium
MY546476529754E
Small
Back

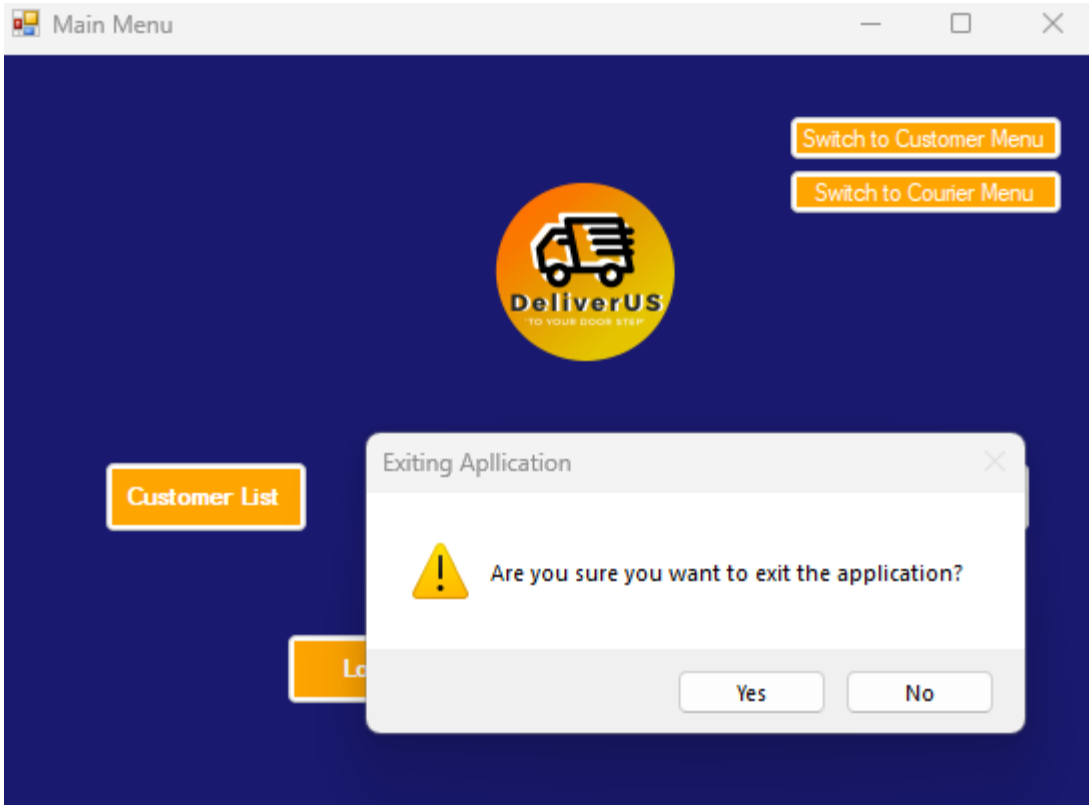

Deliveries
TrackingNumber
Size
CustomerID
ArrivalDate
MY755358866454E
Small
22004123
MY234833124303A
Small
22004123
9/7/2023
MY626254465554A
Small
22004334
5/7/2023
MY456865656545E
Medium
22004123
9/7/2023
MY546476529754E
Small
22001685
7/7/2023
Back


Profile
Full Name:
Eu Kwan Ik
Gender:
Male
Phone Number:
01047808721
Student ID Number:
22001674
E-mail:
xeddpotato@gmail.com
Back
Save


Main Menu
Switch to Customer Menu
Switch to Courier Menu
DeliverUS
Customer List
Courier List
Customer Feedback
Log Out
Exit


Courier List
Student ID
Search
Student ID
Name
Email
Phone
Gender
RegistrationDate
22001674
Eu Kwan Ik
xeddpotato@gmail.com
01047808721
Male
4/7/2023
22001765
Muhammad Badri Bin A...
badriferdinand
0126533275
Male
1/7/2023
Back


Customer Reviews
Review
Rating
Very convinient! No longer ...
5
Easy to use
5
Back


Main Menu
Switch to Customer Menu
Switch to Courier Menu
DeliverUS
Customer List
Exiting Apllication
Are you sure you want to exit the application?
Yes
No